

Inventory & Order Management SaaS

2-Person Team Plan — Backend Pehle, Phir Frontend

Shuru Karne Se Pehle — 3 Zaroori Faislay

Backend shuru karne se pehle dono log baith kar ye teen cheezein decide karein, warna baad mein conflict ho ga:

- **Database design (ERD):** saari tables aur unke relations ek page pe likh lo
- **TenantId strategy:** har table mein TenantId column hoga — ye rule dono ko pata hona chahiye
- **Git repo structure:** ek hi repo. alag-alag branches (feature/products, feature/orders waghera)

Ye 3 cheezein 1 din mein baith kar decide karo, phir kaam split karo.

Backend Split — 2 Persons

Backend ko do logical halves mein todo: ek banda "Foundation + Catalog" karega, doosra "Transactions" karega — kyunke ye dono kaafi had tak independent hain.

Person A — Foundation + Catalog Owner

Week 1 (yeh pehle banana hai, kyunke B isi pe depend karta hai):

- Project setup: ASP.NET Core Web API, folder structure (API, Application, Domain, Infrastructure, Shared)
- Database connection (SQL Server + EF Core)
- TenantId base entity/middleware setup (har table isko inherit karegi)
- Authentication: Register, Login, Logout, Refresh Token, Password Hashing
- Authorization: RBAC roles (Super Admin → Company Admin → Manager → Salesperson → Warehouse Staff → Accountant)
- Multi-tenancy: Organization create karna, organization profile

Week 2 Onwards:

- Users & Permissions module
- Products module (CRUD + variants)
- Categories & Brands module
- Warehouses module
- Audit Logs (har module ke sath judta hai, isliye foundation banda hi banaye)

Person B — Transactions Owner

Week 1 (jab tak A auth/tenant setup kar raha hai):

- Dummy/mock data ke sath apna kaam design karo (entities Person A se discuss karke)
- Customers module (CRUD)
- Suppliers module (CRUD)

Week 2 (jab A ka auth ready ho jaye, integrate karo):

- Inventory module (stock tracking logic)

- Stock Movements (audit trail)
- Stock Adjustments
- Stock Transfers (warehouse to warehouse)

Week 3:

- Sales Orders (poora lifecycle: Draft → Confirmed → Processing → Shipped → Delivered)
- Purchase Orders
- Invoices
- Payments

Week 4:

- Dashboard APIs (summary counts, low stock, sales today)
- Basic Reports APIs

Ye Split Sahi Kyun Hai?

Person A ka kaam (auth, tenant, products, categories) base hai — B ka poora kaam isi pe khada hai (har order/inventory record ko Product aur TenantId chahiye). Isliye A thoda pehle shuru karega, B parallel mein Customers/Suppliers pe kaam karega (unko products ki zaroorat nahi). Jab Product module ready ho jaye, B turant Inventory/Orders pe move kar sakta hai.

Roz ka rule: har roz 15 minute ka call/chat karo — "maine kya banaya, kal kya banaunga, mujhe tumse kya chahiye". Isse dono sync mein rahenge.

Backend Complete Hone Ke Baad — Testing Phase (2-3 din)

- Swagger/Postman collection banao saare endpoints ki
- Dono log ek dusre ke modules test karein
- Seed data dalo (dummy products, customers, orders) taake frontend banate waqt real data mile

Frontend Split — 2 Persons

Frontend mein wahi logic chalega: jisne jo module backend mein banaya, wahi uska frontend banaye (usko APIs sabse zyada pata hain).

Person A — Foundation + Catalog Pages

- Login/Register/Forgot Password pages
- App layout (Sidebar, Navbar, Protected Routes based on role)
- Organization settings page
- Products page (list, create, edit, variants)
- Categories & Brands pages
- Warehouses page
- Users & Roles management page

Person B — Transaction Pages

- Customers & Suppliers pages
- Inventory dashboard (stock levels, low stock alerts)
- Stock Adjustment/Transfer pages

- Sales Order page (sabse complex — order create karna, products add karna, totals calculate karna)
- Purchase Order page
- Invoices & Payments page
- Main Dashboard (charts, summary cards)

Frontend Folder Structure (dono isi mein apna apna feature dalein)

```
src/ ■■■ components/ (shared: Button, Table, Modal) ■■■ pages/ ■■■ features/ ■ ■■■
products/ ← Person A ■ ■■■ categories/ ← Person A ■ ■■■ warehouses/ ←
Person A ■ ■■■ inventory/ ← Person B ■ ■■■ orders/ ← Person B ■ ■■■
customers/ ← Person B ■ ■■■ suppliers/ ← Person B ■■■ services/ (API calls –
.ts files) ■■■ hooks/ ■■■ store/ ■■■ types/ ■■■ utils/
```

Total Timeline (Roughly)

Phase	Kaam	Time
Planning	ERD + rules decide karna	1 din
Backend Week 1	Auth + Tenant + Products (A), Customers/Suppliers (B)	1 week
Backend Week 2-4	Inventory, Orders, Invoices, Reports	3 weeks
Testing	Postman testing, seed data	2-3 din
Frontend	Sath sath dono apne modules	3-4 weeks

Note: Ye poora MVP scope hai (PDF ke "Phase 1" ke mutabiq). Returns, Advanced Pricing, Notifications, aur AI Features — ye Phase 2/3 hain, abhi mat chhero.